OSPE DAY 3

SPOT NO. 1.

1. Write the full names of transporters A, B and C.
   Ans: A → Sodium Glucose Linked Transporter 1 (SGLT1); B → Glucose Transporter 5 (GLUT5); C → Glucose Transporter 2 (GLUT2)
2. Where else is transporter C found in the body?
   Ans: Plasma membrane of liver, pancreatic, intestinal, kidney cells and the hypothalamus
3. By which type of transport is glucose absorbed from intestinal lumen?
   Ans: Secondary active transport

SPOT NO. 2.

1. Name the sphincter shown
   Ans: Sphincter of Oddi
2. What is the effect of the hormone CCK upon it?
   Ans: It relaxes the sphincter
3. Write down the rest of the effects of CCK on GI tract
   Ans: 1. Contraction of Gallbladder; 2. Inhibits gastric emptying; 3. Stimulates acinar cells of pancreas to release a juice rich in pancreatic digestive enzymes

SPOT NO. 3.

1. Write down the voltage b/w which slow wave potential fluctuates.
   Ans: – 65 to – 45 mV
2. Depolarization phase of action potential is due to which ion?
   Ans: Calcium entry into smooth muscle
3. Repolarization phase of action potential is due to which ion?
   Ans: K+ exit from smooth muscle

SPOT NO. 4.

1. Name any 3 hormones / neurotransmitters (A, B, C) that stimulate HCl secretion.
   Ans: Acetylcholine; Gastrin; Histamine
2. What is protein D called?
   Ans: Proton pump or H-K cation pump
3. Write down the name of a common drug used against the protein D
   Ans: Omeprazole or Risek

SPOT NO. 5.

1. Name the type of movement seen in the large intestine
   Ans: Mass movement
2. How many times / day these movements occur?
   Ans: 3 – 4 times / day
3. What is the name of the reflex that initiate this movement?
   Ans: Gastrocolic reflex